

What is Cryptography Hiding from Itself?

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Abstract

The European Commission’s 2022 proposal for a regulation on child sexual abuse material, popularly labelled *ChatControl*, obliges online services to detect, report, and remove prohibited content, through client-side scanning. This paper examines the proposal as a case of *undone science* in computer security ethics: a domain where technical feasibility and rights-compatibility questions remain systematically underexplored. Combining legal analysis with philosophy of technology, the paper argues that client-side scanning transforms end-to-end encryption from a right to secrecy into a conditional privilege of use. By integrating Isaiah Berlin’s concept of negative liberty, Langdon Winner’s account of the politics of artifacts, and David Hess’s notion of undone science, the analysis traces how design choices become moral constraints. The discussion situates the European debate within broader concerns about proportionality, epistemic selectivity, and the governance of digital infrastructures. Ultimately, the study shows that the controversy over ChatControl is not only about privacy or child protection but about the epistemic norms that define what counts as legitimate technological knowledge.

1 Introduction

In 2022, the European Commission proposed a regulation requiring digital service providers to detect, report, and remove online child sexual abuse material (CSAM). The measure, popularly labelled *ChatControl*, would permit or compel client-side scanning of private communications to identify both known and previously unknown material. Presented as a child-protection initiative, the proposal raises fundamental questions about the compatibility of automated detection with encryption, privacy, and proportionality in European law. Rather than assessing the measure solely in technical or legal terms, this paper approaches it as a site where moral reasoning, epistemic assumptions, and design practices intersect.

From a philosophy-of-science perspective, *ChatControl* exemplifies how technical regulation constructs what counts as reliable knowledge about risk and safety. The proposal relies on predictive machine-learning models to decide when a message is suspect, transforming empirical uncertainty into administrative certainty [15]. Such processes illuminate a broader question in the ethics of technology: how infrastructures embed particular epistemic and moral orders, and what forms of inquiry remain undone or marginalized in the process [24, 28]. This conceptual lens situates the debate as a human-rights controversy as well as a case study in the co-production of knowledge and governance.

The following analysis proceeds in three stages. First, it reconstructs the proposal’s technical and institutional context; second, it interprets its moral logic through classic theories of freedom and power; and third, it situates the controversy within the framework of undone science.

Thesis and contributions. This paper advances three interrelated claims. First, it argues that mandatory *client-side scanning* and so-called *detection orders* amount to a form of *general monitoring*, functionally incompatible with end-to-end encryption (E2EE) and therefore at odds with the jurisprudence of the European Court of Human Rights in *Podchasov v. Russia* [13]. Second, drawing on Isaiah Berlin’s distinction between negative and positive liberty [4], Langdon Winner’s notion that artifacts

have politics [44], and Michel Foucault’s analysis of disciplinary power [19], it situates the *ChatControl* debate within the tension between freedom and technological obedience. Third, it outlines policy alternatives, targeted, warrant-based device operations, perpetrator-side disruption, and opt-in child-safety features, that can enhance child protection without instituting general monitoring.

This work contributes to philosophy of science by extending its attention from laboratory and theoretical knowledge to the epistemic infrastructures of regulation. Classical philosophy of science investigates how evidence legitimates scientific claims, while here the focus lies on how technical standards and detection models produce authoritative knowledge of “safety”. By showing how algorithmic measures come to define what counts as reliable or legitimate knowledge, the paper positions itself within an expanded philosophy of science concerned with governance epistemologies. In this sense, the ChatControl proposal offers a case study in the co-production of knowledge and authority, where scientific rationality and regulatory legitimacy become mutually reinforcing. Furthermore, it positions *ChatControl* as an instance of undone science: a field where crucial normative and empirical investigations are deferred in the name of technological urgency.

Before tracing the legislative and technical history of ChatControl, we first turn to the philosophical vocabulary through which questions of freedom, fear, and obedience acquire meaning in the digital age.

2 Freedom, Fear, and Technological Obedience

At the heart of the ChatControl debate lies a struggle between two moral intuitions that have structured political thought since the Enlightenment: the desire for freedom from interference and the longing for freedom from fear. Isaiah Berlin distinguished these as negative and positive liberty. One is the right to be left alone, the other the power to live without oppression or insecurity. In the rhetoric of digital governance, these two ideals often collide. The promise to make the digital sphere “safe” appeals to the second sense of freedom, yet its realization often undermines the first. In practice, what is presented as a gain in safety frequently materializes as a loss of autonomy. The question, then, is not simply how to balance security and freedom, but how to prevent one form of freedom from devouring the other.

Conceptual grounding. Berlin’s seminal essay distinguishes between *negative liberty* (freedom from coercion) and *positive liberty*, the collective power to realize certain ends [4]. In the *ChatControl* proposal, positive liberty is invoked through the rhetoric of safety, while negative liberty is constrained by design as the architecture itself defines what users may do. Winner noted that *artifacts have politics* [44], meaning that technological systems can embed political choices about control and responsibility. Here, mandatory scanning creates what might be called an *architecture of obedience*: users comply not through law but through design. This dynamic also echoes Foucault’s analysis of panoptic power, in which visibility becomes a mechanism of self-discipline [19]. The result is not overt repression but a subtle transformation of agency in the name of protection.

This tension is amplified by what Hannah Arendt called the politics of fear. Modern states justify extraordinary interventions through the specter of threat. Yesterday it was terrorism, today it’s child exploitation. Fear goes beyond an emotion to an organizing principle of governance. When citizens are made to live in fear, they may come to accept a condition of constant supervision as the price of safety. This dynamic recurs throughout the history of surveillance as technological infrastructures built for protection gradually normalize observation and obedience. The moral danger is not only that individuals lose privacy, but that societies lose the capacity to imagine political life without perpetual monitoring. As Arendt warned, the normalization of fear transforms exceptional measures into permanent ones, converting the emotional register of insecurity into a durable principle of administration.

Finally, the idea of technological obedience captures a subtler dimension of this transformation. By embedding surveillance capabilities directly into personal devices, the apparatus of control becomes intimate, routine, and invisible. The technology itself instructs citizens in compliance. The following sections examine how this tension between freedom and protection is negotiated in policy and design.

This tension resonates with broader currents in contemporary philosophy of technology. Verbeek’s mediation theory emphasizes that moral agency is not external to technology but emerges through its design and use, while Feenberg’s critical constructivism highlights the democratic stakes of that design by asking who controls technological rationality [43, 16]. Likewise, recent work in information ethics and science-and-technology studies, from Floridi’s account of informational morality to DeNardis’s analysis of digital governance, underscores that technical infrastructures already embody normative choices about autonomy and responsibility [17, 8]. Situating *ChatControl* within this landscape shows that the question is not whether technology should intervene, but how its architectures distribute moral agency.

3 Historical and Political Context

3.1 Precedents and the Road to ChatControl

The origins of the ChatControl proposal can be traced to growing concerns over the proliferation of child abuse imagery and the perceived “going dark” problem caused by strong encryption. For over a decade, large technology companies have voluntarily used tools like hash-matching (e.g. Microsoft’s PhotoDNA) to detect known CSAM in users’ uploads to cloud services and email. However, these efforts were limited to unencrypted platforms and were not legally mandated. In 2021, the EU enacted a temporary derogation to its privacy laws (dubbed “ChatControl 1.0” by critics) that explicitly allowed online service providers to continue voluntary scanning of communications for CSAM despite the privacy protections in the *ePrivacy Directive* [14, 10]. This interim regulation permitted companies such as Facebook, Google, and others to detect and report illegal images on their services without running afoul of EU privacy rules, but it made such scanning neither obligatory nor applicable to end-to-end encrypted messaging [12]. Crucially, the 2021 law was set to expire by mid-2024, creating a policy crossroads: European lawmakers could either extend the status quo or attempt a more aggressive approach to online child protection.

Against this backdrop, on May 11, 2022, EU Commissioner for Home Affairs Ylva Johansson unveiled a proposal for a comprehensive regulation to “prevent and combat child sexual abuse” online [29]. Johansson framed the issue in dire terms, warning that without new powers Europe could “turn into a pedophiles’ paradise” and that law enforcement was “going blind” in the fight against child abuse due to offenders hiding in private, encrypted channels. The proposed regulation represented a dramatic expansion beyond the voluntary regime. It sought to make content-scanning obligatory for virtually all providers of digital communications in the EU, including encrypted messaging services. The nickname “ChatControl 2.0” soon emerged to emphasize this shift from optional monitoring to government-mandated surveillance of chats, emails, file uploads and more [23]. What had previously been left to companies’ discretion would become a legal duty, fundamentally changing the relationship between user privacy and platform responsibility.

The introduction of the ChatControl proposal immediately set off alarm bells among privacy advocates, cybersecurity experts, and some policymakers. Many drew parallels to past European debates on mass surveillance. They noted that the Court of Justice of the EU had repeatedly struck down indiscriminate data retention schemes as incompatible with fundamental rights, and warned that the new plan to scan all communications could face the same fate. Digital rights organizations and academics quickly mobilized against ChatControl, branding it the largest expansion of surveillance powers in EU history. The colloquial term “ChatControl” itself, initially popularized by critics, underscored the perception that the proposal was less about targeted law enforcement and more about blanket “control” over private correspondence.

3.2 Legislative Developments (2022–2025)

Over the next few years, the ChatControl proposal made its way through the EU’s legislative process amid heated debate and evolving political climate. After the Commission’s 2022 unveiling, the draft

law was scrutinized by the European Parliament and Council, yielding a chorus of critical assessments. The European Data Protection Board and the European Data Protection Supervisor (the EU’s top privacy regulators) issued a joint opinion warning that the proposal could lead to “generalized and indiscriminate scanning” of personal communications, an outcome they deemed disproportionate and legally dubious. The Council of the EU’s own Legal Service echoed these concerns, cautioning that screening every citizen’s communications would infringe the fundamental right to privacy and likely violate the EU Charter of Fundamental Rights and European human rights law. By late 2022, even the United Nations High Commissioner for Human Rights had weighed in, describing aspects of the proposal as incompatible with international human rights standards and tantamount to mass surveillance and censorship [14, 31].

Facing this barrage of criticism, Commissioner Johansson and other supporters defended the proposal as misunderstood, insisting that it was not about mass surveillance but about protecting children. Nonetheless, significant pushback continued through 2023. In an extraordinary move, the European Parliament’s Committee on Civil Liberties (LIBE) commissioned an additional expert impact assessment on the draft law. The resulting study was highly critical. It concluded there were no known technologies that could scan encrypted content for abuse material without an unacceptably high error rate, and it warned that the proposal, as written, would undermine end-to-end encryption and could even flag consensual personal content (for example, teenagers’ intimate messages) as illicit.

Political resistance to indiscriminate scanning solidified. In November 2023, the LIBE Committee voted to reject the idea of general monitoring, instead suggesting that any measures should be limited to targeting specific suspects rather than everyone, and explicitly calling for the protection of secure encryption in the final law. A pivotal moment came in February 2024, when the European Court of Human Rights issued a ruling in an unrelated case that nonetheless sent a clear signal: requiring encryption to be weakened or bypassed for law enforcement access “cannot be regarded as necessary in a democratic society”. This landmark judgment, though arising from a Russian case (the *Podchasov* case about an order to decrypt Telegram messages), reverberated throughout Europe as a strong affirmation that blanket decryption mandates violate fundamental rights [13]. Members of European Parliament such as Patrick Breyer hailed the ECHR’s decision as effectively declaring the Commission’s client-side scanning plans illegal under human rights law [5].

By mid-2024, these developments had thrown the fate of ChatControl into uncertainty. In June 2024, with EU governments deeply divided and public controversy growing, the Council of the EU postponed its scheduled vote on the legislation. Rather than pushing through a text that many believed would be struck down in court or politically unviable, EU leaders opted to take more time. The temporary “ChatControl 1.0” regime of voluntary scanning was quietly extended for an additional two years to ensure no gap in enforcement while negotiations continued[42]. This extension underscored that, contrary to Commissioner Johansson’s earlier warnings, it was indeed possible to maintain some level of CSAM detection without immediately implementing a radical new surveillance system.

However, the proposal was far from dead. In 2025, the debate reignited under new leadership in the Council. Denmark, holding the rotating Presidency of the EU Council in the latter half of 2025, made advancing the CSAM regulation a top priority [11]. A revised draft emerged, with some changes intended to address concerns—for instance, attempts were made to reduce the rebrand or narrow the scanning obligations to visual content¹ and URLs (using terms like “detection orders” and “upload filters”); and to bolster safeguards. Yet the core requirement for providers to scan private communications remained in the text, meaning the fundamental controversy persisted. By late 2025, the Council was once again pressing for an agreement to introduce ChatControl, prompting digital rights groups to intensify their campaigns to “Stop ChatControl” and protect encryption. As of this writing, the legislative process is ongoing. Europe stands at a crossroads, deciding whether to implement this unprecedented surveillance regime or seek alternative solutions. The historical trajectory of ChatControl, from its ideological inception to the present juncture, illustrates both the resilience of the proposal’s supporters and the

¹“which should be understood as images and the visual components of videos, including charts, infographics, logos, animations, iconography, gifs, stickers or the visual components of livestreaming”, as per the proposal.

equally determined resistance from a broad coalition of opponents. This context sets the stage for examining the proposal’s substance, the arguments for and against it, and the larger ethical implications at play.

4 The Technical Proposal: Surveillance by Design

The ChatControl regulation introduces a complex technical framework that would fundamentally reshape how online communications are policed. In essence, it seeks to embed a permanent scanning apparatus into the digital services Europeans use every day, from messaging apps to email providers to cloud storage platforms. This section breaks down the key technical elements of the proposal, including the scope of coverage, the detection technologies proposed (especially in the context of end-to-end encryption), and the compliance mechanisms such as the planned EU Center for Child Sexual Abuse that would coordinate the effort.

Technical definitions. For clarity, this analysis distinguishes four detection mechanisms described in the Commission’s proposal [12, 6]. (i) *Hash-matching* of known CSAM files; (ii) *perceptual or similarity matching* using machine-learning classifiers to detect previously unknown material; (iii) *grooming-detection* algorithms for text and audio²; and (iv) *upload-moderation gateways* that scan content locally before encryption. Only the first approach has proven reliability; the latter three remain probabilistic, risk false positives, and when deployed before encryption, effectively bypass the guarantees of E2EE [1, 26].

4.1 Scope of Mandatory Scanning

The proposed law mandates that online service providers “detect, report, and remove” child sexual abuse material on their services [12]. This obligation would apply across a broad group of digital platforms. Everything from social media and chat apps to email, file-hosting services, and potentially even app stores would be within scope. Notably, the draft regulation explicitly includes *interpersonal communication services* (like messaging and voice/video chat applications) and *storage services* (like cloud backups or photo albums) in its remit. In practical terms, this means that private one-to-one or group communications which traditionally considered confidential, are no exception. Providers would be required to monitor messages, images, videos, and other files exchanged on their platforms for any sign of illicit content.

One significant aspect is that the rules are *technology-neutral* in terms of network architecture, meaning they make no distinction between unencrypted and end-to-end encrypted (E2EE) services. Popular messengers that employ E2EE for security, such as WhatsApp, Signal, and iMessage, would still fall under the regulation’s requirements. The law does not explicitly demand that companies break or weaken their encryption algorithms; instead, it mandates an outcome (detection of illegal content) and leaves it to providers to figure out *how* to achieve this, even in an encrypted environment. In effect, however, this creates pressure to implement some form of client-side scanning or other workaround to inspect content that providers otherwise cannot read. As one analysis bluntly summarized, ChatControl 2.0 would make scanning obligatory “even for encrypted communications – which would mean either content scanning prior to encryption or encryption keys managed by the service provider instead of the end user” [1]. In simpler terms, if the law passes, an E2EE service provider has two main options to comply. Build a scanning mechanism into the user’s device (before messages get encrypted or after they are decrypted), or alter the encryption model so that the provider itself can access content (for example, via a form of key escrow or by inserting a privileged “ghost” receiver in conversations).

The scope of content to be detected under the proposal is expansive. Providers must look not only for *known* CSAM (for instance, images or videos that have been previously identified and hashed into

²Grooming detection was removed from the Danish presidency versions of the proposal, but the intention to include it back through a future review clause persists.

databases of illegal content) but also for *new or previously unseen* material that could be CSAM, and even for textual patterns that suggest *grooming of minors for sexual purposes*. Detecting known CSAM can be done via cryptographic hash matching using databases of known illegal images (such databases are maintained by organizations like NCMEC in the US or Interpol). The far thornier task is identifying new CSAM content and grooming conversations, which likely requires machine learning classifiers and artificial intelligence. The draft regulation anticipates the use of AI-driven tools to scan images for likely abuse (even if not identical to known illegal files) and to analyze message text for signs of an adult engaging in sexual predation of a child. These “classifiers” raise complicated questions about accuracy, training data, and the risk of false positives, which we will explore in the critique section. But from a technical proposal standpoint, the law essentially demands that providers employ whatever emerging technology can serve these detection aims. A provider could be ordered, for example, to install an AI model that flags nude images of minors not seen before, or to use natural language processing on chat transcripts to flag grooming language.

The breadth of services covered is tempered slightly by some exemptions and graduated obligations. Small enterprises and non-profit platforms feared that they would be unable to implement such sophisticated monitoring. The proposal’s text indicates that only *commercial* services are obligated, and it carves out an exemption for services that are “non-commercial and not public-facing” or that do not rely on advertising to fund operations [12]. This suggests that some open-source, decentralized, or hobbyist services might not be directly forced to comply if they have no revenue model. However, the line can be blurry. For instance, an open-source encrypted app that is free but listed on commercial app stores might still be pressured via app store age restrictions (another aspect of the proposal). The regulation also introduces the concept of **risk assessments**: providers would have to periodically assess and report the risk that their service is used for CSAM dissemination or grooming. If a service is deemed low-risk, broad scanning mandates might not immediately apply. Conversely, if a significant risk is identified (for example, a chat platform known to be used by predators), authorities could issue a *detection order* compelling that service to implement approved detection measures. In theory, this approach targets interventions where needed; in practice, critics suspect nearly all major communication platforms will end up receiving such orders, given the universal potential for abuse on any platform. Indeed, privacy advocates point out that even a platform that has no known problem could be labeled “at risk” simply because its strong encryption prevents anyone from knowing whether abuse is present or not. This is a Catch-22 that would result in nearly all secure services being forced to scan.

Another piece of the proposal is *age verification and age-based access controls*. The law includes provisions to restrict minors from accessing services that do not implement the required scanning or other child safety measures. It suggests that app stores and service providers should verify users’ ages or otherwise block users under 17 from using communication tools that could expose them to unfiltered content [12]. In effect, if an app chooses not to build scanning capabilities (for instance, an encrypted messaging app that refuses to comply), it could be forced off the mainstream market for underage users, or even entirely, if regulators interpret offering an un-scanned service as non-compliance. This introduces a potential requirement for invasive age verification techniques (since determining who is a minor is non-trivial online) and raises further privacy issues, as age verification often entails collecting personal data or IDs from users. The inclusion of age verification highlights the comprehensive approach of the proposal which not only seeks to scan content, but it also aims to limit who can use which services, in the name of shielding children from harm.

To coordinate and work through all these obligations, the regulation would establish a new EU Center on Child Sexual Abuse. This Center would act as central coordination hub. It would maintain databases of known CSAM (hash lists and possibly feature patterns for known abuse content), develop and approve detection tools, receive reports of flagged content from companies, and then forward validated reports to the appropriate national law enforcement agencies. The EU Center is envisioned as providing expert support and ensuring consistent standards and supporters say it will prevent the chaos of each company or country doing its own thing. It would also issue guidance on best practices and be the contact point for service providers who need to implement the law’s requirements. By

centralizing expertise and data, the Center hopes to streamline the handling of CSAM reports and improve cross-border cooperation (since currently, a report of abuse content in one country often must be relayed to law enforcement in another country where the content is hosted or the user resides, causing delays). The Center would effectively be the eyes of the system, reviewing the output of automated scans and deciding what gets passed to police for investigation.

In summary, the technical blueprint of ChatControl is sweeping. If enacted, it would oblige providers big and small to build detection mechanisms into their services, covering all types of content and communications, even those secured by encryption. It envisions harnessing advanced algorithms and an EU-wide institutional infrastructure to continuously monitor and intercept prohibited material. The next subsection examines more closely how this can (or cannot) be reconciled with end-to-end encryption, which is at the heart of the technical controversy.

4.2 Client-Side Scanning and the Encryption Dilemma

Because so many modern communication services employ end-to-end encryption, the ChatControl proposal's demand for content scanning raises a fundamental technical challenge: how can you scan the contents of a message that is encrypted such that only the sender and recipient are supposed to see it? End-to-end encryption (E2EE) means that service providers do not have the decryption keys; they literally cannot read user messages in transit or on their servers. This is celebrated as a crucial security and privacy feature—preventing hackers, governments, or even the service operators themselves from snooping on conversations—but it directly conflicts with any regime of automated content inspection. The EU proposal, by being “technology-neutral”, implicitly pressures companies to find some method to square this circle without explicitly mandating them to backdoor their encryption. The most commonly floated solution is what's known as *client-side scanning* (CSS).

Client-side scanning involves moving the detection process to the user's device (the client) rather than on the server. In an E2EE scenario, this typically means that before a message is encrypted and sent out (or right after it is decrypted on the receiving end), an automated scan takes place on the device. If the scan finds content that matches known illegal material or is deemed suspicious, it can then notify authorities or the provider, even while the message remains encrypted in transit. Essentially, CSS turns your phone or computer into a content police officer that checks your data locally and sends an alert if something illicit is found. Under ChatControl, this could manifest as a runtime monitoring service on your phone that continuously checks your outgoing messages and media against the CSAM hash database and AI classifiers provided (via the EU Center or your service provider).

The concept of client-side scanning shot to global prominence in 2021 when Apple announced plans to deploy a CSAM detection system on iPhones. Apple's proposal was to have iPhones compare users' photos against a database of known CSAM hashes *before* those photos were uploaded to iCloud; if a certain threshold of matches was found, Apple would be alerted and could pass information to authorities. The plan sparked immediate backlash from security researchers and privacy advocates who argued that such a system would effectively make every iPhone perform on-device surveillance, opening the door to abuse and eroding user trust in device security. After widespread opposition, Apple reversed course and shelved its client-side scanning feature by the end of 2021, acknowledging that it could not implement the system in a way that preserved user privacy to the standard the company wanted [34]. This episode was closely watched in Europe. It demonstrated both that a major tech company found the idea feasible to implement technically, and that even a well-resourced, privacy-conscious company ultimately concluded the trade-offs were too great and the technology not mature enough to deploy safely [1].

Despite Apple's retreat, client-side scanning remains essentially the only currently imagined way to enforce ChatControl on encrypted services without outright breaking the encryption protocols. EU officials have sometimes suggested that client-side scanning is not equivalent to breaking encryption, a claim that has been met with skepticism and ridicule in the tech community. For instance, documents from the Danish Presidency of the Council claimed that client-side scanning “does not break end-to-end encryption” because the encryption process itself remains intact [36]. Technically, it's true the encrypted

channel isn't being backdoored in the traditional sense; however, from a security perspective, installing scanning software on a device has a very similar effect to a backdoor. It means a third party's code is examining content that was supposed to be private to the user. Security experts liken it to putting "spyware" on everyone's phone with government approval [32]. If the government (or law) can dictate that an app must scan your messages on your phone, the end result is that the privacy of the encryption is nullified as one of the "ends" of the communication (your device) has been effectively deputized as an informant. As the Electronic Frontier Foundation put it, client-side scanning inherently *undermines* the promise of E2EE because a communication cannot be considered secure if one of the endpoints is systematically reporting content to authorities [32].

There are alternative approaches sometimes discussed, such as *server-side scanning with key escrow* (where companies keep a copy of encryption keys or maintain the ability to decrypt messages on the server side). This, however, is just a more overt way of breaking encryption and creates huge security risks (because any key escrow or backdoor can be discovered and misused by malicious actors). The ChatControl proposal avoids openly endorsing this approach, likely because it would blatantly violate commitments not to outlaw encryption. Instead, the focus is on compelling companies to achieve the "functional equivalent" of a backdoor via client-side scanning. From a user's point of view, both methods result in their private messages being inspected; from a hacker's point of view, either method introduces new vulnerabilities (a treasure trove of keys to steal, or a scanning algorithm on devices that could potentially be tampered with).

Implementing client-side scanning in practice raises many technical challenges. One is ensuring the scanning algorithms themselves are accurate and secure. They would need to be updated with hash lists and AI models, likely delivered from the central EU Center to user devices. Keeping those confidential is important (to prevent adversaries from figuring out how to evade detection by studying the model), but distributing them widely to millions of devices inherently risks leaks. Another challenge is device compatibility and performance. Older smartphones or computers may struggle with running intensive image classification on-device for every photo, for example. Then there is the user experience issue: will users be notified if their content is flagged? (The current designs suggest not; the scanning is meant to be silent to avoid tipping off criminals). This secretive aspect can lead to false positives remaining unchallenged by the user until law enforcement perhaps knocks on the door.

The regulation's text suggests that detection orders for E2EE services should specify the least privacy-intrusive measures and could, for instance, be limited to targeting specific types of content (e.g. only known CSAM images, or only grooming text, etc.). The intention is to show that it's not an indiscriminate wiretap; rather it is searching only for illegal digital content. However, the technical reality is that to search for illegal content, one must *examine everything*. Even if the algorithm is trained to ignore unrelated content, it still has to analyze all messages or images to decide whether they match the forbidden patterns or not. Thus, any client-side scanner will be, by design, an ever-present surveillance program on the device, inspecting all content before sending or upon receiving. This blurs the line between a limited search and a general one.

In summary, the ChatControl proposal forces a reckoning with the encryption dilemma: to comply, encrypted service providers will likely resort to client-side scanning as the only currently feasible method. This approach means inserting content monitoring software into the previously private sphere of the user's device. The next sections of this paper will examine the arguments from supporters who contend that such measures are justified and can be implemented responsibly, as well as the counterarguments from critics who see this as a bridge too far technically, practically, and ethically.

4.3 The Epistemology of Safety

Beyond its technical framework, the ChatControl proposal illustrates how "safety" becomes an epistemic construct. Within regulatory discourse, safety is treated as a measurable property, as something that can be detected, verified, and optimized, rather than as a moral judgment requiring deliberation. This transformation converts the political question "what harms are intolerable?" into the technical question "what signals exceed a detection threshold?". As Sheila Jasanoff observes, modern governance often

stabilizes uncertainty through “technologies of hubris”, translating moral risk into calculable indicators of control [27]. In this sense, client-side scanning performs an epistemological shift because it redefines protection as the ability to classify. What counts as a safe or unsafe communication becomes a product of algorithmic confidence rather than public reasoning. Recognizing this transformation is essential, because it frames the later debate not only as a conflict over encryption or privacy, but over who is entitled to define knowledge of harm in a digital society.

5 Supporters’ Rationale and Motivations

The ChatControl proposal has been championed by a coalition of policymakers, child protection organizations, and law enforcement officials who argue that the gravity and scale of online child sexual abuse demand strong action. This section outlines the key points made by supporters of the regulation. They emphasize the alarming statistics of abuse, the inadequacy of current voluntary measures, the anticipated benefits of a centralized EU approach, and the safeguards they believe can reconcile the proposal with fundamental rights. By understanding the supporters’ perspective, one can appreciate the earnest motivations driving the legislation, even as one weighs them against the serious concerns.

5.1 Protecting Children and Closing the Gaps

First and foremost, advocates of ChatControl stress the moral imperative of protecting children from sexual exploitation and abuse. They point to an explosion of reported online abuse cases in recent years as evidence that existing efforts are not keeping pace with the threat. EU Commissioner Johansson often cites a staggering “6000% increase” in reports of child sexual abuse online in the EU over the last decade [12, 29]. According to data shared by the Commission, the U.S. National Center for Missing and Exploited Children (NCMEC) received around 85 million files flagged as potential CSAM in 2021 alone [18]. This was up from 65 million the year before, indicating an accelerating problem. Moreover, experts estimate that this may be only “the tip of the iceberg” of abuse content: countless images and victims remain hidden, unreported, or undetected under the current regime [12]. These numbers are cited to impress upon lawmakers and the public that online child abuse is a crisis of epic proportions that is growing with each passing year.

Supporters argue that the status quo approach, which relies heavily on U.S.-based hotlines and the goodwill of tech companies, has left dangerous gaps. They note that over 60% of known CSAM worldwide is actually hosted on servers in Europe [20], yet European law enforcement still depends on NCMEC and American tech platforms to flag most of it. This reliance on voluntary action and transatlantic cooperation can lead to delays and blind spots. The European Commission’s Directorate-General for Home Affairs has pointed out that some service providers choose not to participate in voluntary detection programs at all, meaning abuse on those platforms goes unchecked. Even among those who do scan, companies can change their policies at any time if there’s no legal requirement, potentially creating “safe havens” where offenders will flock if a major service stops detecting abuse. Proponents find it unacceptable that a perpetrator’s ability to abuse children online might simply hinge on which app or provider they use. The clear solution, in their view, is to make detection and reporting mandatory across the board, so that there are no dark corners of the mainstream internet where abusers can operate with impunity.

A related rationale is the so-called “going dark” problem [2]. Criminals increasingly use encryption to shield their activities from law enforcement. Police and child-protection units testify that their investigations are often hindered by encrypted devices or chats; vital evidence of abuse is inaccessible due to strong encryption. Supporters of the EU proposal argue that society must not allow encryption to become an absolute sanctuary for child predators. As Johansson bluntly put it, without intervention, abusers will exploit secure communications and “our justice system will go blind” to their crimes. The ChatControl legislation is framed as a necessary evolution of law enforcement capabilities where there is an update to the tools and legal mandates to match the reality that harmful activity has moved to

encrypted platforms. Many supporters acknowledge encryption's benefits but contend that it cannot be a justification for inaction when children's safety is at stake. In their moral reasoning, the prevention of horrific crimes against children outweighs the abstract privacy interest in keeping all communications off-limits to any scanning. They often invoke the visceral ethical line that if scanning could rescue even one child from ongoing abuse or prevent the viral spread of a single horrific video of a child's rape, then it is warranted. From this perspective, opposing surveillance measures can seem tantamount to prioritizing abusers' secrecy over victims' suffering.

Supporters also highlight specific qualitative examples to bolster their case. They speak of instances where abuse was only discovered because an electronic service did flag CSAM, thereby saving a child from an active situation of molestation. They worry that if encryption becomes an impenetrable shield, such children would remain trapped in abuse for lack of any reporting trail. Additionally, the pro-regulation camp underscores that perpetrators are not just hypothetical. Law enforcement agencies in Europe are inundated with tens of thousands of leads annually (mostly forwarded by NCMEC from companies like Facebook) and are struggling to investigate them all. The sheer volume is overwhelming under current approaches; supporters argue that a more systematic, EU-wide scanning and coordination system could help manage this load by filtering and prioritizing cases through the proposed EU Center.

In public communications, proponents often use strong rhetorical appeals. Campaigns in favor of the law have featured testimonies of abuse survivors and their advocates, emphasizing the lifelong trauma caused by these crimes and the urgency of doing more to stop online exploitation. The Brussels-based NGO Brave Movement and other survivor-led groups have been vocal, arguing that privacy arguments should not drown out the voices of victims. Johansson and allied ministers frequently invoke the phrase "think of the children" not as a mere slogan but as a genuine moral appeal. They assert that as policymakers, they have a duty to err on the side of saving children from abuse, even if it means challenging some established norms on privacy.

Notably, a coalition of child protection organizations welcomed the Commission's proposal in an open letter, calling it a "timely and historic" step in the global fight against child sexual abuse [38]. This letter, co-signed in May 2022 by groups including the Internet Watch Foundation (IWF), Thorn (a U.S.-based anti-exploitation NGO), ECPAT, and others, provides a clear window into the supporters' mindset. It acknowledges the scale of the problem and praises the EU for taking leadership. The letter highlights that current legislative frameworks vary across EU member states and rely on voluntary action by providers, which is insufficient and inconsistent [38]. The regulation, the groups say, offers a "harmonised, binding, and future-proof framework" to tackle not only known CSAM but also new content and grooming, anticipating evolving technologies and tactics of abusers [38]. Importantly, these child-focused organizations assert that the proposal aims to ensure technology is used in conformity with fundamental rights, including the rights of children to be protected and not subjected to sexual violence [38]. In other words, they argue that children's right to safety is as much a fundamental right as adult privacy, and the regulation is an attempt to balance the two.

5.2 Centralization, Transparency, and Safeguards

Another pillar of the supporters' rationale is the creation of the EU Center and the promise of greater transparency and control in how online abuse is tackled. Currently, much of the detection of CSAM is handled within private companies and via a U.S.-centric reporting channel (NCMEC). European authorities often have a limited view into what companies are doing or not doing, and there is no single European entity aggregating data or expertise on this issue. The Commission's proposal argues that establishing a centralized EU Child Sexual Abuse Center will significantly improve the status quo. By being the dedicated hub for all reports, the Center would simplify cooperation—service providers would have one place to send detected CSAM reports, and that Center would then take on the burden of forwarding to the relevant national police and ensuring any needed support (like identifying the child victim or providing hash data) is in place. This, supporters claim, avoids the scenario where providers must figure out jurisdiction and contacts for dozens of different countries' police forces every time they find illegal content.

Centralization is also touted as a means to achieve more **transparency and accountability** in the process. Under voluntary schemes, companies publish only limited transparency reports about their CSAM detection efforts, and there is little external oversight of the tools they use. The EU Center, however, is intended to work under legal frameworks and would presumably issue annual transparency reports about the numbers of detections, the tools used, error rates, etc. Proponents argue that this can actually *increase* trust in the system by subjecting it to public scrutiny and audits. The Center would collect data for transparency reporting and support audits of the detection technologies, as outlined in the proposal. In theory, this means that if a tool proves to be inaccurately flagging too much legitimate content, it would be documented and could be reevaluated or improved. The Commission has emphasized that a goal of the regulation is to prevent “the unintended removal of legitimate content” and address concerns about misuse of the detection tools. Supporters frequently mention that there will be strict safeguards: detection orders would be issued by courts or independent authorities, not arbitrarily by police; the tools and indicators must be “sufficiently reliable” and approved; humans at the EU Center would presumably verify flagged material before any user is subject to investigation, to reduce false alarms.

One often cited safeguard in the draft law is that any use of detection technology must be limited to the purpose of finding CSAM or grooming and *cannot* be repurposed for general law enforcement or other content (like copyright or political speech) [12]. Commissioner Johansson has repeatedly stressed this point: the text explicitly prohibits using the CSAM-scanning tools for anything else and mandates that only “verified indicators” of child sexual abuse provided by the EU Center can be used [12]. This is meant to alleviate fears of mission creep—that once the scanning infrastructure exists, it would be used to spy on unrelated matters. By enshrining in law that the tools are only for CSAM and under oversight, supporters claim they are building a firewall against abuse of the system ³.

The idea of being “technology-neutral but rights-respective” is central to the supporters’ narrative. Johansson has argued that the proposal gets “the balance right” between different fundamental rights [29]. She contends that while privacy is important, it is not absolute and must be balanced with children’s right not to be sexually abused [6]. In her view, the regulation is about finding that equilibrium—obliging companies to prevent or detect abuse as a “last resort”, but doing so under conditions that respect privacy as much as possible. For example, supporters note that the proposal underwent a thorough impact assessment and was accompanied by two years of consultations with stakeholders from both privacy and child rights sectors before it was finalized [12]. This, they argue, shows that the Commission did its due diligence, hearing out critiques and trying to design the law carefully. Indeed, Johansson has listed numerous privacy and security organizations (including EDRI, the Electronic Frontier Foundation, and various encrypted service providers) that her team consulted during drafting, to demonstrate that the concerns were not ignored [12]. The fact that those groups remain opposed does not negate that their input may have shaped certain safeguards.

Supporters also point to public opinion in favor of stronger measures. The Commission cites surveys such as a Eurobarometer in which reportedly 80% of Europeans support stronger rules to fight child abuse online [7]. Another figure mentioned is that 95% of EU citizens say it’s important to have laws forcing providers to act against child sexual abuse [7]. These numbers are used by proponents to suggest that democratic sentiment is on their side, and that privacy activists are out of touch with the average person’s priorities. Of course, the framing of survey questions can influence such high percentages, but politically it gives cover to those pushing the law that they have a mandate from the public to do so.

Among member state governments and MEPs, those in favor of ChatControl often belong to political groups emphasizing law-and-order or child welfare. They argue that law enforcement has been left “fighting crime with 20th-century tools in a 21st-century digital world” and that this regulation modernizes the legal toolkit. Several governments, such as those of Ireland, Spain, and some Eastern European countries, have been generally supportive, noting their police forces’ struggles with cross-border online abuse cases. Law enforcement agencies like Europol are quietly supportive of more data

³Critics are not entirely convinced by this promise, as discussed later, but it remains a key talking point for the proposal’s defenders.

access as well. Although Europol itself has warned about certain implementations, it certainly welcomes more reports and data, as seen by its calls for access to even non-CSAM data discovered (Europol has expressed interest in any images found that might help in other investigations, for instance).

In summary, the supporters' rationale hinges on a few key assertions:

- The scale of online child sexual abuse is enormous and growing, making it imperative to act decisively (backed by statistics of millions of reports and steep increases).
- Voluntary measures and fragmented national laws have left dangerous gaps; only a uniform, mandatory EU-wide framework can ensure no safe haven for abusers ("ending the impunity loopholes").
- The proposed system, centered on an EU Center, will actually improve effectiveness and transparency by coordinating efforts, applying uniform standards, and enabling oversight and accountability of the detection process.
- It is possible to implement scanning in a way that respects fundamental rights: by narrowly focusing on CSAM/grooming, employing safeguards like independent authorization of detection orders, limiting data use to that purpose, and forbidding any broader surveillance applications.
- Protecting children is itself a fundamental rights obligation (as per the rights of the child in EU law), and the proposal strives to balance this with privacy rather than simply steamrolling privacy.
- Public opinion and moral duty favor taking action—doing nothing in the face of preventable child abuse is politically and ethically untenable.

These arguments set the stage for why many well-intentioned actors support ChatControl. However, as the next section will detail, each of these points is met with counterarguments highlighting potential flaws, risks, or counterproductive outcomes of the proposal. The tension between these two sides forms the crux of the ChatControl controversy.

6 Critique and Opposition

From the moment of its announcement, the ChatControl proposal has provoked intense criticism from a wide range of actors. Privacy advocates, cybersecurity experts, legal scholars, and even some government bodies have raised alarms about the implications of mandating mass scanning of communications. Their critiques cover multiple dimensions: fundamental rights violations, technical infeasibility and security dangers, the likelihood of errors and overreach, and broader societal and ethical consequences. In this section, we examine the main points of contention, supported by evidence and expert opinions that question the wisdom of the ChatControl approach.

6.1 Privacy, Fundamental Rights, and Legality

Perhaps the most foundational criticism is that ChatControl is incompatible with the fundamental rights to privacy and confidentiality of communications enshrined in European law. Opponents argue that ordering the indiscriminate scanning of all citizens' private messages is effectively equivalent to a general mass surveillance regime, something that liberal democracies have long rejected as an unacceptable intrusion by the state. The European Union's own legal framework reflects this with past judgments by the Court of Justice of the EU having struck down laws requiring bulk retention of communications data (such as phone metadata) on the grounds that they constituted a disproportionate interference with privacy and data protection rights. Critics see ChatControl as violating the same principles, only in an even more extreme fashion by potentially inspecting actual content of communications, not just metadata. The European Data Protection Supervisor and European Data Protection Board's joint

opinion on the proposal explicitly warned that it may lead to a “de facto generalised and indiscriminate” surveillance of communications, which would likely fail the necessity and proportionality tests required under the EU Charter of Fundamental Rights.

The confidentiality of correspondence which is protected by Article 7 of the Charter and Article 8 of the European Convention, has long been a cornerstone of European privacy. ChatControl reverses that principle by introducing a permanent monitor that treats every citizen as a potential suspect. Civil society groups like European Digital Rights (EDRi) and the Electronic Frontier Foundation have branded ChatControl as “mass surveillance” plain and simple [10], arguing that no matter how well-intentioned, such blanket monitoring is a hallmark of authoritarian regimes, not open democracies.

The UN Human Rights Council’s special experts echoed these concerns in early critiques, noting that the EU proposal may lead to forms of pervasive oversight and automated content control, incompatible with international human rights standards [31]. European human rights officials worry about the chilling effects. If people know their every private message could be analyzed by an algorithm (and possibly seen by authorities), they might self-censor or avoid communicating sensitive, personal, or dissenting information at all. This could undermine freedoms of expression and association, not to mention the trust users have in digital services.

Importantly, in February 2024, the European Court of Human Rights (ECHR) weighed in indirectly via the *Podchasov v. Russia* case. The ECHR struck down a Russian law that had required an encrypted messaging service to assist in decrypting communications for the Federal Security Service, finding that such a broad requirement to make encrypted chats accessible to authorities “cannot be regarded as necessary in a democratic society” [13]. The court explicitly noted that laws enabling general direct access to communications or mandating decryption without strict safeguards violate the right to privacy [13]. Though this case was against Russia, its principles apply to all Council of Europe members (which includes all EU states). Privacy advocates seized on this judgment as essentially a death knell for ChatControl’s core idea: if requiring decryption is illegal under human rights law, then requiring client-side scanning (which is akin to preemptively decrypting by the user’s device) should likewise be seen as disproportionate. Patrick Breyer, a Pirate Party MEP and leading opponent of ChatControl, hailed the ECHR ruling as a “landmark judgment” that clearly establishes the illegality of measures like client-side scanning, which he argues would amount to surveillance of everyone instead of investigation of suspects [5].

Critics also question the legal basis of the ChatControl proposal in relation to other EU laws. They point out that the European Electronic Communications Code and the ePrivacy Directive generally forbid the kind of general monitoring of content that ChatControl would mandate. There is also potential conflict with the Digital Services Act (DSA), which does not impose general monitoring obligations on platforms for illegal content, reflecting a deliberate policy choice to avoid undermining privacy and free speech. For the CSAM proposal to carve out an exception and actually require monitoring is seen as setting a dangerous precedent. The German Parliament’s research services and Digital Affairs Committee, for instance, examined the proposal and found that it likely clashes with the German constitution’s protections for the privacy of correspondence and could contravene the EU’s own GDPR and Charter. Ella Jakubowska of EDRi told the German Bundestag that ChatControl would likely violate the EU Charter of Fundamental Rights as well as the GDPR and the newly minted DSA [9].

Another facet of privacy is the protection of personal data. Scanning content means processing personal data about possibly hundreds of millions of innocent people. Under GDPR, processing must have a lawful basis and be necessary and proportionate to a legitimate aim. Opponents argue that bulk scanning fails these tests, since the vast majority of people scanned will have no involvement in CSAM, and less intrusive means of achieving the aim (targeted investigations) exist. The European Council’s own Legal Service flagged that the general screening of communications “affects the fundamental right to respect for private life as well as the right to protection of personal data” in a severe way. The prospect of sensitive personal communications (which could include health information, intimate photos, political or religious discussions) being analyzed by algorithms and potentially ending up flagged and reviewed by authorities, is seen as an unacceptable intrusion, even if the intention is to

detect criminal material.

In summary, from a fundamental rights standpoint, the critique is that ChatControl is disproportionate and likely unlawful. It treats every citizen's private communication as subject to inspection, reversing decades of privacy protections. The ECHR decision underscores that democratic societies draw the line at such pervasive surveillance. For many, this principle alone should be decisive. No matter how noble the goal, a free society cannot institute a system that amounts to a generalized digital "wiretap" on its population.

From a deontological perspective, generalized scanning fails not due to its ineffectiveness but because it violates the moral constraint of confidential correspondence. Some means are impermissible even when pursued for noble ends; a rights-respecting order cannot institutionalize suspicion of everyone to police a few.

6.2 Security and Technical Feasibility

Experts in cryptography and computer security have been some of the most vocal critics of ChatControl, arguing that the proposal is built on shaky technological grounds and could actively harm cybersecurity for everyone. The primary issue is that any system to scan encrypted content inevitably involves weakening or bypassing encryption, which in turn undermines the security that encryption provides against malicious actors.

One common refrain is: "There's no such thing as a backdoor that only the good guys can use". If a vulnerability or access mechanism is introduced (be it client-side scanning or an encryption backdoor), it creates an opening that determined adversaries could exploit. For instance, if client-side scanning software runs on users' devices, malware authors might target that software to either disable it (if they want to avoid detection) or subvert it (perhaps to plant false evidence or extract data). Security researchers point out that increasing the complexity of devices with such scanning capabilities also increases the attack surface for hackers. The very existence of a database of illegal content fingerprints distributed to devices worldwide is a security risk—if the database leaks, it could reveal what is being detected and help criminals evade detection (or conversely, if someone could inject hashes into the database, they might cause totally benign content to be flagged on all devices).

Additionally, by effectively breaking end-to-end encryption's promise, the proposal could make users more vulnerable to other threats. The Federal Bureau of Investigation (FBI) in the U.S., ironically, has even warned that strong encryption is vital for security in the face of cyber threats; as one example, the FBI recently recommended that Americans use end-to-end encrypted messaging apps to thwart Chinese hackers. If the EU mandates undermining encryption, it could expose European citizens and governments to espionage or crime.

The fact that recent Council drafts propose excluding non-public communication services such as those used for national-security, governmental, or professional secrecy purposes, from the regulation's scope [5] has drawn criticism from privacy advocates. Although the text justifies this carve-out by citing confidentiality and classified-information concerns, opponents argue that it effectively shields state and institutional communications while ordinary citizens' private channels remain subject to scanning. This perceived asymmetry reinforces the view that those in power recognize the risks of such surveillance yet still impose it on the general public, a double standard that has become a rallying point for critics.

The feasibility of accurately detecting only bad content is another point of skepticism. As noted earlier, no AI or classifier is perfect. A leaked study by the European Parliament's research service concluded that no existing technology can detect CSAM without producing error rates so high that large amounts of lawful communication would be wrongly flagged. The Irish police statistics often cited by critics illustrate the problem: in one period, out of 4,192 reports from electronic service providers, only 852 (about 20%) turned out to be actual abuse material; 471 (11%) were confirmed false positives (flagged as abuse but actually innocent content). This suggests that automated filters produce a significant number of mistakes. At the scale of current E2EE deployments, millions of false positives could occur daily, each potentially leading to an innocent person's data being scrutinized by authorities or even

wrongful investigation. These are real people traumatized by being mistakenly accused of heinous behavior (the next subsection on false positives will detail this further).

Computer security specialists also emphasize that the complexity of implementing client-side scanning on all devices, across countless platforms, in a secure manner is extremely high. Some doubt it can be done at all without introducing new flaws. When Apple, one of the world’s most valuable tech companies with top-notch engineers, tried to design such a system, they ultimately abandoned it precisely because they couldn’t guarantee it would only ever be used as intended and wouldn’t erode user privacy [34]. If Apple couldn’t find a way to square that circle, critics ask, why does the EU think forcing every company to try will end any better? There’s also the pragmatic consideration: smaller companies and open-source projects simply don’t have the capability to develop and maintain such sophisticated scanning infrastructure. They might shut down or withdraw from markets instead (we’ve already heard Signal say it would sooner leave the EU than comply [41]). The end result could be consolidation of the tech industry with only a few giants managing to implement scanning, thus further entrenching their dominance, while nimble secure alternatives disappear or go underground. This runs counter to EU’s other policy goals about encouraging competition and digital sovereignty.

Moreover, technical critics underscore the adaptive behavior of criminals. If ChatControl were implemented, savvy abusers would likely adapt by using tools that evade detection. They might encrypt files in ways that break known hashes or use slang/code words to avoid text-based detection. They could switch to decentralized or self-hosted communication methods beyond regulators’ reach. Essentially, the sophisticated offenders (the ones law enforcement most wants to catch) would find ways to remain hidden, while casual users and petty criminals get caught by the driftnet of mass surveillance. Thus, the efficacy of the approach is questioned. Will it truly make a serious dent in CSA crime, or will it mostly generate a flood of low-value alerts (while driving the worst activity further underground)? The Global Encryption Coalition, a group of cybersecurity and human rights organizations, stated that “[ChatControl] will not solve the problem of the online spread of child sexual abuse material, but will introduce significant security risks for all citizens, companies, and governments”[21].

Finally, one recurring theme is that pushing this proposal ignores the counsel of technical experts. The tech community’s near-unanimous opposition is evidenced by open letters signed by hundreds of computer scientists and cryptographers warning that the idea is dangerous and based on “unrealistic ideas”. For instance, hundreds of researchers and academics signed a letter raising the alarm about ChatControl’s consequences. When so many of the world’s leading experts in relevant fields are saying “this won’t work as intended and will make things worse”, critics argue that policymakers have a responsibility to heed those warnings rather than dismiss them. Ignoring them risks repeating mistakes of the past when poorly conceived technology mandates have backfired or been rendered obsolete by clever workarounds.

In summary, the technical critique of ChatControl is that it is not just deeply challenging, but potentially counterproductive. It weakens the security of digital infrastructure, doesn’t reliably identify illicit material without huge collateral damage, and can be circumvented by determined criminals. The proposal resembles installing “government-sanctioned spyware” on devices and it makes mass surveillance almost an inevitability by giving authorities a tool they will inevitably seek to expand. The next part of the critique delves into what those expansions and errors might mean in human terms.

6.3 False Positives, Overreach, and Unintended Consequences

One of the most concrete concerns about ChatControl is the risk of false positives—innocent content or behavior being misidentified as illegal—and the harmful consequences that can result. No detection system is perfect. When applied on a population of hundreds of millions of users sending billions of messages, even a tiny false positive rate could produce an avalanche of wrongful flags. Each false flag could mean a user’s private files are scrutinized by a government analyst, an account gets suspended, or in the worst case, an innocent person is investigated for child abuse. Such events are not hypothetical fear mongering; they have already occurred under the current voluntary systems, and critics warn they would increase dramatically under an expanded mandatory regime.

A widely cited example is the case of “Mark”, a father in the United States whose Google account was terminated and who was reported to the police after he took photos of his toddler’s infected groin area to send to a pediatrician for medical advice [25]. Google’s automated CSAM detection (which uses hash matching and AI) flagged the images as child pornography. Despite the benign context (a parent helping a child get medical care), Google locked the father’s accounts and filed a report with NCMEC, leading to a police investigation. Although police quickly cleared him of wrongdoing, Google refused to reinstate his services, essentially punishing him as if he were a criminal [22, 25]. This real incident, reported by *The New York Times* and *The Guardian*, highlights a few issues. First, algorithms lack context: they saw a naked child image and automatically assumed abuse, whereas a human would understand context matters (a rash photo for a doctor is not exploitation). Second, once a company or authority acts on a flag, it can be very hard for the affected individual to appeal or reverse the damage, as seen by Mark losing years of personal data tied to his Google account with no recourse [25]. Third, it shows how proxies for law enforcement put ordinary people at risk of being swept up by the power of the state due to automated mistakes.

Now imagine scaling this up across Europe. With ChatControl, not only known CSAM images (like in Google’s system) but also new images, videos, and even text chats would be scanned by fallible algorithms. The European Parliament’s study warned of high error rates for detecting new material. False positives could include: parents’ innocent photos of their kids in the bath, teens sharing consensual intimate pictures with each other, artists’ drawings or memes that an AI misclassifies, or text discussions about sexuality that are non-exploitative but flag certain keywords. Each false alert could send a person’s private messages to the new EU Center, and likely onward to law enforcement for review. Even if eventually cleared, that person may have unknowingly had very sensitive information read by government agencies. This erodes trust and may violate privacy in cases where there was no crime at all.

Furthermore, children themselves could be harmed by false positives or overreach. Ironically, a law intended to protect children might end up incriminating them. Consider that a significant proportion of so-called “CSAM” reports involve self-generated images by minors (like teens sexting each other) or exploration that, while concerning, is not a result of adult exploitation. If the system flags a consensual exchange between two minors, it could result in them being caught up in the justice system or facing school discipline, etc. There are instances in some countries where teenagers have been charged with child pornography for possessing images of themselves or classmates. Critics fear ChatControl will multiply such cases unless very carefully handled, essentially criminalizing normal adolescent behavior and violating minors’ privacy rather than protecting them.

Another consequence is the overloading of law enforcement with trivial or irrelevant reports. Law enforcement agencies have limited resources; flooding them with false leads or marginal cases could divert attention from the truly urgent cases of children in real danger. This is what some police representatives have also quietly expressed concern about. Quality over quantity of reports is crucial. If everything becomes noise, the real signals might be missed which is a scenario actually bad for child protection.

Critics also stress the slippery slope of *function creep*. Once the infrastructure for scanning every message is in place, what’s to stop its use from expanding to other content or crimes in the future? History offers many examples where surveillance tools introduced for one purpose got extended: e.g. anti-terror laws applied to ordinary criminal cases, or data retention initially meant for serious crime being accessed for minor offenses. In the context of ChatControl, while the law might initially restrict use to CSAM and grooming, future EU legislatures (or perhaps a determined executive action) might broaden it to scan for terrorism-related communications, hate speech, or other illegal content. Indeed, the technology review one reference alluded to Chinese chat surveillance for political dissent—an extreme scenario, but it shows the tech could be repurposed to hunt for whatever authorities consider dangerous. Opponents argue that installing a general monitoring capability is a Rubicon meaning that once crossed, you have fundamentally altered the citizen-state power balance, and you rely on the perpetual benevolence and restraint of future governments not to misuse it. Given that even in the EU,

there have been abuses (for example, the Pegasus spyware scandal where some governments spied on political opponents [39]), there is reason to doubt such tools would remain limited.

There is also an economic and social consequence to consider. If services like WhatsApp or Signal withdraw from the EU to avoid compliance, Europeans might lose access to some of the most secure and user-friendly communication tools. They could be forced onto less secure platforms (ironically increasing risk of data breaches and other crime) or to more fragmented, possibly subscription-based services. Smaller European companies could be crushed by compliance costs, concentrating even more power in Big Tech (who alone can afford the infrastructure). Additionally, journalists, whistleblowers, and activists in Europe might find it much harder to communicate safely. Organizations like Reporters Without Borders have warned that undermining encryption and introducing surveillance will endanger journalistic sources and investigative journalism, which often rely on secure, confidential channels to expose wrongdoing [40].

A subtler unintended effect is the potential normalization of surveillance. If people grow up or become accustomed to the idea that all their communications are automatically checked by authorities, this could have a chilling effect on free expression and the open exchange of ideas. Self-censorship could become widespread. It might also foster complacency. Citizens might feel less responsibility to be vigilant (for example, parents might think the system will catch any grooming of their kids, so they pay less attention, which might not be true).

Critics argue that there are far better ways to protect children that don't require turning everyone's phone into a sensor for the state. By focusing so much on a technological silver bullet, society risks neglecting measures like education, psychological treatment for offenders, undercover operations, and other targeted interventions that could actually reduce harm without such collateral damage. The next section (Alternatives) will elaborate on those.

In summation of the critique: ChatControl's opponents see it as a well-intentioned but dangerously misguided proposal. Legally, it's on thin ice; technically, it's fraught with peril; practically, it could do as much harm as good. They urge that the EU should not implement a solution that might be worse than the problem—especially not when the stakes involve both the safety of children and the digital rights of an entire society.

7 Alternatives and Solutions

Critics of ChatControl are often asked: "If not this, then what? How else can we tackle the scourge of child sexual abuse online?" It's a fair question. Everyone agrees that protecting children is paramount; the disagreement lies in the methods. Opponents of the current proposal generally assert that there are more proportionate, effective, and rights-respecting strategies to combat online child abuse. In this section, we outline several alternative approaches that have been proposed. These range from more targeted law enforcement tactics to preventive measures and technological innovations that do not compromise privacy to the same extent. The goal is to illustrate that rejecting ChatControl is not the same as doing nothing about the problem at hand. It is about choosing smarter interventions that uphold fundamental values while addressing the problem.

7.1 Targeted Investigations and Judicial Warrants

One alternative approach is to refocus on *targeted surveillance of suspects* rather than blanket scanning of the general population. This means using traditional investigative techniques and obtaining judicial warrants to monitor individuals when there is reasonable suspicion of involvement in child abuse offenses. The idea is that law enforcement should pursue the criminals, not cast a net over everyone. For example, police can infiltrate child exploitation networks, run undercover stings, or trace leads from existing CSAM distribution points, and then—when they identify a suspect—apply for a court order to surveil that suspect's communications or search their devices. This aligns with how most law enforcement operates for other serious crimes. Specific evidence triggers targeted surveillance or

searches under judicial oversight.

The European Parliament's LIBE committee, in its amendments, essentially endorsed this concept by voting in late 2023 to remove indiscriminate chat control and instead allow *targeted* monitoring of individuals or groups for whom there is a justified suspicion. They also emphasized protecting encryption in the law, reflecting the idea that you can go after perpetrators without weakening the security of everyone else's tools. This approach would avoid the mass privacy intrusion of scanning every message by limiting interference to those cases where there is cause. It also leverages the expertise of investigators to discern context and focus on high-value targets, something an automated system cannot do.

To make targeted investigations effective, law enforcement agencies could be given more resources and training to handle digital evidence, trace cryptocurrency transactions (often how child abuse content is paid for), and collaborate across borders swiftly. International collaboration is key here as much of the child abuse material found in Europe comes from or is circulated via other continents. Strengthening partnerships with entities like NCMEC, Interpol, and police in other countries can lead to more heads of networks being identified and arrested, without necessarily scanning every European's chats. In fact, recent busts of large child abuse platforms (like the takedown of the darknet site "Boystown" by German police) were achieved through classic police work that involves undercover agents, informants, and international coordination, not mass surveillance of communications.

When targeted surveillance is used, it should be under strict judicial control and oversight to ensure it remains proportionate. European courts have signaled that while general dragnets are illegal, targeted data access can be permissible for serious crimes if authorized properly (the ECHR and CJEU have allowed targeted wiretaps or data preservation orders). By confining the intrusion to where it's needed, you maintain the principle of individualized suspicion that underlies democratic policing. Importantly, this also means innocent people's data largely remains untouched, law enforcement is not inundated with noise, and the rights of the vast majority are not curtailed.

Some might argue that a targeted-only approach misses incidents where the perpetrator wasn't previously suspected. However, this is where other measures come in (like maintaining cooperation with tech companies to continue their voluntary scanning in non-encrypted spaces, which is currently the case). An example of a balanced strategy follows. Continue scanning on platforms where it is feasible (e.g., cloud storage, social networks like Facebook where scanning of images has been happening and can continue under appropriate legal basis), but do not force it into E2EE spaces. The voluntary scanning regime already yields an enormous number of reports (those 85 million files to NCMEC in 2021), more than law enforcement can fully process. It might be more fruitful to better process and act on the existing reports than to expand the net further. In effect, improve the "last mile" – the actual rescue of children and arrest of abusers based on the data we already have – rather than exponentially increase data volume.

7.2 Strengthening Existing Measures and International Cooperation

Many opponents of ChatControl suggest that the EU could achieve significant gains by bolstering and refining the existing toolkit against child abuse rather than introducing a radically new surveillance paradigm. Here are some measures often proposed:

1. Enhanced Cooperation with Service Providers (Voluntary Efforts 2.0): Instead of mandating scanning in E2EE, the EU could work with tech companies to ensure they are doing everything they reasonably can in non-E2EE contexts and that they promptly act on user reports or other signals of abuse. Many large platforms already have trust-and-safety teams; increasing transparency and perhaps standards for how they moderate and remove CSAM can help. The EU could, for instance, require annual transparency reports from platforms on their CSAM detection and removal efforts (something the DSA may indirectly foster as well). If some smaller or foreign platforms are lagging, diplomatic or legal pressure (short of breaking encryption) could be applied to bring them into cooperative frameworks. Remember that the 2021 derogation (ChatControl 1.0) was voluntary; it could be possible to extend that regime with improvements – for example, providing EU funding or technical support for smaller

companies to implement hashing tools, or creating EU-based reporting channels so that not everything has to route through NCMEC in the USA (one aspect of the EU Center idea that could be preserved without scanning E2EE).

2. Takedown and Blocking of Hosting Sites: A lot of CSAM is traded on image hosting sites or dark web forums that are not end-to-end encrypted platforms but specialized repositories. The EU has been working on a parallel track of legislation (a Child Sexual Abuse Directive) to expedite the removal of CSAM from websites and servers. Prioritizing that – basically, cutting off the supply at the source – could have big impact. If 62% of CSAM is hosted in Europe[12], then empowering Europol and national police to quickly seize servers or compel hosts to delete content (with due process) can help. The EU can also collaborate with other countries to take down major hubs of content. This is more a classic cybercrime approach where the sites where illegal content resides are identified and eliminated, rather than chasing every copy in every private message.

3. Improved Cross-Border Law Enforcement Cooperation: One rationale the Commission gave for ChatControl was that currently Europe relies on the US for reports, and cases get delayed. A good alternative solution is to bolster EU’s own law enforcement coordination and capabilities. Europol’s role could be expanded with more officers dedicated to child sexual abuse cases, improving the speed and thoroughness of investigations that begin with a report. The creation of an EU Center could still happen *in a limited form*: it might act as a central hub for reports from companies (as they voluntarily or under existing law scan public or stored content) and a hub for police cooperation. If detection orders are off the table, the Center could focus on supporting national units with analysis and victim identification expertise. This would achieve some benefits of centralization (expertise, avoiding duplication, building better forensic tools to identify victims from images, etc.) without necessarily scanning E2EE.

4. Empowering Hotlines and NGOs: Europe has a network of Internet hotlines (like INHOPE member hotlines) where the public can report CSAM they stumble across. Strengthening these hotlines, funding them better, and integrating their work with law enforcement can help remove content quickly and find perpetrators. Also, supporting NGOs that operate in this space (like those that do “web scraping” for open CSAM or that work on victim support) can indirectly reduce harm and improve enforcement.

5. Focus on Offender Prevention and Deterrence: Another angle is to prevent the creation of new CSAM and the commission of abuse in the first place. This could involve public awareness campaigns, better mental health support and intervention for at-risk individuals (there are organizations that work on rehabilitating or treating people with pedophilic disorders to prevent offending), and tougher deterrent sentences for those who do offend (though punishment alone is not a panacea, clear legal consequences do matter). If fewer crimes occur, less abuse content will be produced and circulated. Similarly, continuing efforts to physically bust networks (like international operations to arrest child abuse rings) can have a huge impact – e.g., taking down a major site or ring can free many children from ongoing abuse and also remove a source of new CSAM.

6. Privacy-Preserving Technologies Research: The EU could invest in research into technologies that might enable some level of protection without the downsides of current scanning. For instance, exploring the use of cryptographic techniques like homomorphic encryption or secure enclaves to detect illegal material in an encrypted environment without exposing everything. As of now, these are mostly theoretical or not practical at scale, but funding research could yield future breakthroughs. Importantly, such research should be transparent and involve the security community (the kind of open dialogue Apple belatedly engaged in, which helped identify the issues in their approach). If a way could be found to, say, match encrypted images to known CSAM hashes without revealing anything about non-matching images, that would be a game-changer. But until then, it might be prudent to hold off on mandates and instead channel resources into developing better tools.

7.3 Education, Empowerment, and Prevention

Beyond law enforcement and tech measures, many experts highlight the importance of education and empowerment strategies to combat child exploitation. After all, purely technical solutions address the

symptom (the content) after the fact, whereas preventive approaches aim to stop abuse from happening or escalating.

One key alternative approach is *digital education for children and parents*. Teaching children from an early age about online risks, how to recognize grooming behavior, and how to seek help can make them more resilient against predatory tactics. Often, abusers manipulate or coerce minors into sharing images or into in-person meetings. If children are equipped to spot warning signs and know that they can talk to trusted adults, some exploitation can be averted. Similarly, educating parents and teachers to understand how grooming occurs online (through gaming platforms, chats, social media) enables them to better protect and supervise without overbearing surveillance. Many child rights advocates argue for a focus on “empowerment of children, not exclusion” – meaning rather than simply banning kids from platforms or surveilling them, give them the knowledge and tools to navigate safely.

Another aspect is *victim support and reporting mechanisms*. Encouraging children or teens who experience or witness exploitation to come forward and ensuring they have safe, confidential channels to do so (like helplines, school counselors, etc.) can lead to intervention in cases that no algorithm might catch (for example, if abuse is happening in person or on platforms that aren’t scanned). Some survivors of abuse might only reveal it if they trust the system will help them rather than punish or shame them. Thus, strengthening child protective services and NGOs that support victims is crucial.

When it comes to minors sharing self-generated explicit content, an approach advocated by some is harm reduction. Educating teens on the risks, encouraging a culture where sharing someone else’s intimate images without consent is socially unacceptable (thus reducing distribution), and perhaps employing device-level parental controls (that families opt into) that can detect when a minor is about to send a sensitive image and warn them. Notably, this latter idea was part of Apple’s plan – they implemented a feature (separate from CSAM scanning) where iMessage could detect nudity in images and warn under-18 users and optionally notify parents. This kind of feature is client-side but under user/parent control, not mandated by government and not reporting to law enforcement. It is a more consensual tool that some parents might choose to enable to protect their kids, as opposed to an imposed system. Encouraging development of such family safety tools, while preserving the voluntary aspect, is an alternative path.

The *ethical use of AI* in moderation can still play a role, but perhaps on the client or family side rather than the state side. For instance, there could be AI that helps identify when an online conversation is heading in a risky direction (grooming) and either alerts the child (“Be careful, this person might not be who you think”) or alerts a parent if guardianship settings allow. This again would ideally be opt-in. It’s akin to driving a car with seatbelts and airbags – provide protective technology to users, rather than mandate that the police monitor every driver all the time for dangerous behavior.

At the societal level, tackling root causes of abuse is an ultimate preventive measure. That goes beyond the scope of internet policy, but things like combating poverty, supporting families, improving mental health services, and cracking down on organized crime that facilitates trafficking, all contribute to reducing child sexual abuse. The online dimension is just one part of a complex issue.

An important alternative perspective raised by some experts, including at the EDPS, is to reframe the question that is being asked. Rather than “how can we scan more to find abusers”, ask “why are so many children being abused in the first place and what can be done offline to stop it?”. Most abuse material originates from real abuse of a child. If we invest in preventing that abuse or identifying it through social services and community awareness, we reduce the material at the source. For instance, training teachers and doctors to spot signs of abuse could lead to more rescues without requiring mass surveillance of the internet.

Finally, when considering alternatives, many opponents of ChatControl say that one alternative is simply *not to do the things that clearly undermine security* and wait for more evidence-driven policies. The EU could extend the current voluntary regime (as it has to 2025/26) and use that time to gather data. Which approaches yield the best outcomes? Are there maybe hybrid models (e.g., scanning only metadata or scanning only files above a certain risk score)? Are there legal tweaks (like clarifying how companies can share data with police under GDPR while still respecting privacy) that can help?

Essentially, pause and study rather than rush into a permanent system. This sentiment was echoed by many MEPs who opposed rushing the law in 2023/24. In policy, sometimes doing nothing drastic is better than doing something harmful. Meanwhile, incremental improvements in law enforcement cooperation and the other measures listed can proceed.

In conclusion, there is a rich menu of alternative actions: rigorous, targeted policing with warrants; enhanced cooperation and takedown of content at sources; supporting victims and witnesses; educating and empowering youth; and innovating on safety tools that don't violate privacy. Taken together, these strategies may not be as politically dramatic as declaring "we'll scan everything to catch predators", but they might actually be more effective in reducing harm. Importantly, they uphold the values of a free society, ensuring that in protecting children we do not create a surveillance state that could harm everyone's liberties.

8 Philosophical Foundations

To critically understand the EU's client-side scanning proposal for combating CSAM, we turn to three seminal thinkers—Isaiah Berlin, Langdon Winner, and Michel Foucault—whose work on liberty, technology, and power illuminates a deeper shift in digital governance from protection to surveillance. Each provides a conceptual lens on how a tool intended to safeguard society can transform into an instrument of control. This section examines how Berlin's account of liberty, Winner's thesis that artifacts have politics, and Foucault's notion of disciplinary power together frame the philosophical foundations of the debate. In doing so, it elucidates how a well-meaning protective measure can invert into a pervasive surveillance regime, setting the stage for the ethical and societal analysis that follows.

8.1 Berlin: Liberty and the Paradox of Protection

Isaiah Berlin's classic distinction between negative and positive liberty [4] helps clarify the value conflicts at the heart of client-side scanning. Negative liberty denotes freedom from interference—an individual's right to privacy and to be left alone by authority. Positive liberty, by contrast, aspires to freedom to pursue one's own potential or collective good, even if it sometimes invites constraints in service of that goal. The EU's proposed scanning mandate ostensibly serves a positive aim which is to protect children and society (a collective good) by proactively detecting abuse material. Yet it achieves this by interfering with personal communications, thereby encroaching on negative liberty. Berlin warned that the logic of positive liberty, in the wrong hands, can become a rationale for coercion and authoritarianism [4]. In the name of some higher moral objective or "true" freedom, authorities may claim the right to override individuals' explicit freedoms "for their own good". This inversion of liberty (using the language of collective freedom to justify intrusive control) resonates uneasily with the client-side scanning paradigm. What begins as a protective venture (freeing society from the harm of CSAM) risks sliding into a blanket surveillance of citizens, treating privacy and personal autonomy as secondary. Berlin's framework thus highlights a tragic trade-off. The very freedom from fear and harm promised by positive security measures can erode the freedom from intrusion that defines liberal society. His insights sound a cautionary note that even well-intentioned digital safety policies must be checked by liberal principles, so that the project of protection does not collapse into a perversion of the very liberty it claims to defend, imposing overarching control in the name of freedom.

8.2 Winner: Technology as Politics by Other Means

If Berlin establishes what values are at stake, Langdon Winner's thesis that "artifacts have politics" [44] explains how choosing a technological design is effectively choosing a form of governance. Winner argues that technical systems are not neutral tools but they embody particular arrangements of power and authority. In his words, technologies can be understood as "forms of order": their architectures predispose certain social and political outcomes. Client-side scanning can be seen as an example of this phenomenon. By mandating the installation of scanning algorithms within personal devices and

encrypted services, policymakers are inscribing a political choice into technology. That is the choice to prioritize centralized oversight and control over distributed privacy. The architecture of scanning software inherently shifts the balance of power because it creates an infrastructure where platform providers (and by extension the state or regulators) gain an always-on inspection capability, while users lose a degree of control over their information. In Winner's terms, this technical arrangement encodes a particular vision of order. An order in which surveillance is normalized as a built-in feature of digital life. Notably, this shift is self-reinforcing. Once the artifact of client-side scanning is in place, it can silently extend to other objectives (today CSAM, tomorrow perhaps hate speech or political dissidence), much as Winner observed that the political consequences of a technology often exceed its initial intent. The very design that enables child protection can be repurposed to broader social monitoring, illustrating Winner's point that who gets to design and control technical systems will implicitly decide who gets controlled. By viewing the EU's proposal through Winner's lens, we see it as a decision to embed a permanent political function into technology. In short, client-side scanning turns a moral mandate ("protect children online") into a governance infrastructure. Into a transformation of policy into code that blurs the line between safety features and a surveillance apparatus.

8.3 Foucault: Surveillance, Discipline, and Digital Panopticism

Michel Foucault's account of disciplinary power offers a third perspective, explaining the social effects of embedding surveillance into everyday life. Foucault used Bentham's Panopticon as a metaphor for modern regimes of monitoring and control [19]. In a panoptic system, individuals know they may be watched at any moment, even if they are not continuously being observed. This uncertainty (a state of conscious and permanent visibility) leads people to internalize the surveillance and regulate themselves, yielding what Foucault describes as the automatic functioning of power. Client-side scanning can be seen as a digital-era instantiation of the panoptic principle. By automating the inspection of every message or image against forbidden content, it positions each personal device as a potential surveillance node. Users, aware that their private communications are subject to machine scrutiny, may begin to self-censor and adjust their online behavior preemptively. Even though the primary target is illegal content like CSAM, the net cast by ubiquitous scanning envelops all users in a system of observation. Foucault's insight was that the effectiveness of such power lies in its subtlety where overt coercion is replaced by a diffuse, continuous surveillance that people come to accept and even internalize as normal. In the context of the EU's proposal, what starts as a protective measure against criminals takes on the character of disciplinary power applied to the population at large. The logic of prevention demands that everyone's data be scanned, effectively treating every citizen as a potential suspect who must demonstrate innocence. This marks a shift from a liberal model of governance, where power intervenes after a wrongdoing, to a disciplinary model, where power seeks to prevent wrongdoing by constant monitoring and normalization. The transformation from protection to surveillance is thus illuminated by Foucault's framework: the technology intended to secure society simultaneously serves to survey and discipline it. Seeing ChatControl through Foucault's lens, client-side scanning helps construct a digital environment in which individuals are subtly nudged to align with prescribed norms (in this case, absolute abstention from illicit content), under the unblinking gaze of algorithmic inspectors. The result is a governance paradigm that blurs care and control like a parent who both protects and perpetually watches, eroding the distinction between nurturing guidance and constant oversight.

8.4 From Safeguarding to Surveillance

Taken together, the ideas of Berlin, Winner, and Foucault reveal how a policy like client-side scanning, born from the imperative to safeguard children, can evolve into a far-reaching reconfiguration of liberty and power in the digital sphere. Berlin teaches us that any attempt to enforce collective "good" (security, moral hygiene) through intrusive means must reckon with the loss of personal freedom it entails. Winner reminds us that implementing such a measure is a technical restructuring of the public-private boundary to one that embeds a particular power dynamic (state or corporate supervision over individual

communication) into the fabric of technology. Foucault, finally, uncovers the potential consequences for society: the emergence of a surveillance-driven order in which compliance is silently compelled and individuals adjust to being watched. What we witness in the EU’s CSAM proposal is emblematic of a broader philosophical shift in digital governance. The ethos of protection (grounded in safeguarding rights and welfare) is increasingly entangled with the practices of surveillance, which manage risk by pervasive oversight. These foundational perspectives urge us to critically question whether the cure (permanent scanning of our digital lives) might corrupt the values it seeks to defend. They underscore that client-side scanning is a decision about what kind of digital society we want, and how we balance the twin imperatives of security and liberty.

In conclusion, this philosophical inquiry provides a deeper context for evaluating client-side scanning. It shows that the proposal sits at the intersection of competing liberties, encodes political choices in technical form, and could usher in subtle new modes of disciplinary power. Recognizing these dimensions is essential before embracing technologies that “protect” by watching over us. With these insights in mind, we now transition to the Ethical and Societal Considerations of client-side scanning, examining in concrete terms the risks, benefits, and moral trade-offs that attend this shift from a protective ideal to a surveillance reality.

9 Ethical and Societal Considerations

The battle over ChatControl is not only a technical or legal one; at its core, it is also an ethical debate. It exemplifies an instrumental view of technology. Moral problems are recast as engineering problems to be solved by design. The risk is that technical means then drive ends, quietly redefining “safety” as continuous inspection and “freedom” as conditional permission. It forces us to confront questions about the kind of society we want to live in, how we value privacy versus security, and how far we are willing to go to pursue even very important goals. This section will discuss some of the key ethical and societal considerations in this debate, including the tension between collective safety and individual rights, the implications for trust in technology and institutions, and the precedent that adopting such a law could set domestically and internationally.

Moral framework. From an ethical standpoint, the proposal juxtaposes two incompatible evaluative logics. Its defenders reason largely in consequentialist terms: if scanning prevents even one instance of child abuse, the outcome justifies the intrusion. Opponents adopt a deontological view, maintaining that certain rights—privacy, confidentiality of correspondence, and freedom of expression—possess an inviolable character and cannot be overridden by aggregate benefits [10]. This clash mirrors a long philosophical tension between outcome-based and duty-based ethics [30, 33]. Yet in practice, the two are not symmetrical. The utilitarian logic depends on empirical success, whereas the deontological constraint defines the legitimate boundaries of policy experimentation. Clarifying which moral vocabulary is operative in legislative reasoning is therefore essential for assessing the legitimacy of surveillance-based safety measures.

9.1 Rights, Liberties, and the Social Contract

One ethical lens to apply is a rights-based approach. From this perspective, certain rights — like the right to privacy, freedom of expression, and due process — are fundamental and should not be overridden except in the most extreme and justified circumstances. Privacy is an essential component of dignity, autonomy, and intimacy, it’s not just about hiding wrongdoing. It allows people to explore ideas, develop relationships, and express themselves freely without fear of constant judgment or reprisal. Mass surveillance, by its very nature, threatens to erode this private sphere that is crucial for individual freedom.

Ethicists often cite the famous maxim attributed to Benjamin Franklin: “Those who would give up essential liberty to purchase a little temporary safety deserve neither liberty nor safety”. While

protecting children is more than just “a little safety”, the spirit of the quote is that some liberties are so essential that sacrificing them may ultimately leave society worse off, even if the intention was good. If we create a system where every message and call can be monitored, we risk normalizing an intrusion into personal lives that fundamentally shifts the power between state and citizen. Today it may be to find child abusers, but tomorrow could it be to find political dissidents? The *slippery slope* concern is real in ethical terms. Once a tool of control is in place, even if instituted with noble intent, it can be repurposed or abused by others later. European history, with its chapters of authoritarian rule and secret police surveillance (from the Gestapo and Stasi to more modern examples), provides a haunting backdrop. The slogan “Never again” in the context of fascism has often been linked to ensuring that states cannot intrude so deeply into private lives.

There is also the principle of *proportionality* in ethics and law — that measures taken by authorities should be appropriate and not excessive relative to the problem. Critics argue that ChatControl is a disproportional response. It treats the entire population as suspects to potentially catch a tiny fraction of criminals. This is analogous to doing house-to-house searches of everyone’s home in the country on the chance of finding a few criminal stashes. Democracies abhor such general searches (the Fourth Amendment of the U.S. Constitution, for example, was a reaction to general warrants used by colonial British authorities). In Europe, general monitoring is seen in a similar light. The ethical underpinning is that people have a right to a private life unless there’s an individualized reason to invade it. By breaking that norm, ChatControl raises the question: are we willing to rewrite the social contract, trading the presumption of privacy for a promise of possibly catching more bad guys?

Supporters might argue that protecting children is a moral duty that can override other considerations — a utilitarian view that if surveillance prevents severe harm to even a single child, it could justify inconvenience or even loss of privacy for others. This gets into classic ethical debates of *utilitarianism* vs. *deontological ethics*. Utilitarianism focuses on outcomes (maximizing overall well-being, even if some rights are infringed), whereas deontological ethics focuses on duties and principles (certain actions are wrong in themselves, even if they lead to good outcomes, because they violate rights or moral rules). ChatControl proponents take a somewhat utilitarian stance: preventing child abuse is such a weighty outcome that it justifies extreme means. Opponents often take a deontological stance: scanning everyone’s private communications is inherently wrong, a form of collective surveillance that should not be done, no matter the intention.

Strikingly, both sides invoke protecting the innocent as an ethical priority. One side says “think of the innocent children harmed by abuse”, the other says “think of the innocent people whose privacy and lives will be harmed by false suspicion and constant surveillance”. Ethically, a balance must be found that minimizes harm to both groups of innocents. The challenge is that the two goods (child safety and privacy) seem to be in conflict here. But perhaps they are not as zero-sum as presented; as some ethicists point out, pitting privacy versus safety is a false dichotomy when smart policy could enhance both (for example, targeted policing can protect children while leaving the general population’s privacy intact).

9.2 Trust in Technology and Governance

Another ethical and societal aspect is the *erosion of trust*. Modern societies heavily rely on trust in both technology and institutions. If people come to feel that their devices are effectively “spying” on them, they may lose confidence in using digital services for legitimate purposes. End-to-end encryption has been a technological trust guarantee — an assurance that your private conversation really is private. Breaking that (via client-side scanning) could lead citizens to avoid using digital communications for anything sensitive, or to seek out underground tools to re-establish privacy. That could ironically fragment the internet and push some communications into harder-to-monitor channels anyway (which might defeat the intended purpose, as criminals also adopt more secure methods).

Trust in government is also at stake. The EU enjoys relatively high trust among certain populations partly because it’s seen as upholding rights and rule of law. If the EU were to implement a system widely perceived as “Big Brother” surveillance, it could fuel public resentment, especially if and when

the first stories of abuses or mistakes come out (imagine headlines of an elderly person mistakenly raided due to an algorithm’s error, or a data breach of the CSAM hash database causing chaos). Already, we’ve seen youth and privacy activists mobilizing (petitions, protests) against ChatControl, suggesting that a portion of the population sees this as an illegitimate overreach.

There is also the hypocrisy factor damaging trust. The revelation that EU officials wanted to exempt themselves from scanning [5] suggests a “rules for thee but not for me” approach. Ethically, for laws to have legitimacy, they should apply equally to all (with very limited exceptions perhaps for national security communications, but even that is contentious). Exempting the powerful implicitly acknowledges that the surveillance poses risks, yet deems those risks acceptable for the general public. The epistemic gap mirrors the digital privacy divide, referring to the unequal access to privacy knowledge and tools: those imbued with the authority to define privacy are not affected as much as those lacking the epistemic power to contest them. The consequence is manifesting into existence yet another layer of the divide, now institutionalizing unequal access to privacy through bureaucratic privileges.

If trust in institutions declines, the effectiveness of those institutions in protecting children could also decline. People might be less willing to report abuse if they fear the information will be mishandled or used to surveil them. Or they might shun cooperating with law enforcement, viewing them as part of an oppressive surveillance apparatus. The social cohesion can fray.

On the flip side, one could argue that failing to act also erodes trust in the safety of the digital environment for children. If nothing is done, and abuse continues to rise, that could also undermine trust in the state’s ability to provide security. So, policymakers have to navigate which approach preserves overall trust better. The ethical argument from critics is that an approach respectful of rights preserves trust in the long run better than one that might create a security state.

9.3 Global Precedent and the Authoritarian Drift

Europe is often looked to as a leader in human rights and digital regulation. If the EU were to adopt ChatControl, it could set a precedent that other countries follow. Some of those countries might be less scrupulous about limiting the surveillance to just CSAM. An authoritarian regime could point to the EU and say: “We too are implementing on-device scanning of encrypted messages — the EU does it for CSAM, we will do it for terrorism or subversion”. The existence of the technology and legal blueprint lowers the barrier for authoritarian abuse. Meredith Whittaker of Signal pointed out that the UK’s flirtation with scanning in its Online Safety Bill, had it gone through, would have set a “permissive international precedent” for mass surveillance, and that stopping it was huge to arrest that momentum [41]. The same applies to the EU. It’s a significant chunk of the world. If it normalizes mass client-side scanning, it could become “standard practice” globally in a way that makes the internet uniformly surveilled.

This concern was echoed by the EFF and many NGOs in the global encryption coalition: they worry about a domino effect. We already saw how Australia’s 2018 anti-encryption law (requiring companies to assist in access to encrypted data) was cited by other governments to justify their own moves, and how tech companies resisted in one jurisdiction but then had to consider what to do when multiple governments demand similar access [35]. For example, Apple’s now-cancelled CSAM scanning plan was intended for the US, but they faced immediate questions about what they would do if other governments asked to expand it to other content. Apple’s retreat was partly because they couldn’t convincingly say it would never be misused for other purposes.

The ethical stance for many is that Europe should hold the line and not betray its values. Essentially, “do no harm” in the sense of not pioneering a system that could be co-opted for repression. If even Europe — with its strong rights frameworks — says yes to mass scanning, it legitimizes the concept worldwide.

Finally, there is an ethical dimension regarding *children’s rights themselves*. Children have a right to privacy too. While protecting them from abuse is critical, treating all children’s communications as suspect could also be seen as a breach of their rights. For instance, teenagers might be deterred from seeking advice about sensitive issues (sexuality, mental health, etc.) if they think their messages

might be read by some adult authority. Also, if a minor shares something foolish or intimate and it gets caught in this system, the trauma of that exposure could also be harmful. So one must consider not just abstract adult privacy vs child safety, but children's own privacy vs their safety. The UN Convention on the Rights of the Child talks about both protection from harm and rights to privacy and expression. An ethical approach tries to respect the whole range of a child's rights. A mass surveillance approach might be too one-sided on protection at the expense of other aspects of their well-being, autonomy, and trust.

In sum, the ethical debate around ChatControl reveals it as a clash between deeply held values: the desire to protect the vulnerable at all costs, and the commitment to civil liberties that define an open society. It raises questions of what kind of future we want. One where technology is bent towards surveillance and control, or one where it empowers individuals and requires law enforcement to do the hard work within constraints. While everyone supports the end goal of reducing child abuse, the means cannot be seen in isolation from the moral context. In moral terms, the challenge is not to weigh privacy against child safety as rival goods, but to determine whether certain forms of protection are permissible means to those ends. Policies that condition safety on continuous surveillance risk transforming moral responsibility into technical compliance. A sustainable ethic of digital governance would instead treat rights not as obstacles to protection but as constitutive of it, viewing them as the very conditions that make trust, communication, and accountability possible [3, 4, 37].

By carefully weighing these ethical considerations, legislators are challenged to find solutions that protect children without forfeiting the principles of privacy and freedom. The hope of many critics is that Europe can craft a path that addresses the problem through targeted, innovative measures and ethical governance — setting a positive precedent that protecting children and preserving rights are not mutually exclusive, but mutually reinforcing in a just society.

A further worry is the drift from emergency to normalcy. Fear and administrative rationality can harden temporary exceptions into permanent infrastructures. If inspection becomes routine then citizens learn to self-censor, and public freedom contracts by design.

10 Conclusion

The debate over the EU's ChatControl proposal illuminates a fundamental tension at the heart of modern governance: how to safeguard the most vulnerable among us without sacrificing the rights and freedoms that protect us all. Over the course of this analysis, we have explored the historical genesis of the proposal, its technical blueprint, the arguments marshaled by its proponents, the extensive critiques raised by its opponents, possible alternative strategies, and the broader ethical stakes involved.

ChatControl was conceived in response to a very real and urgent problem. Few issues evoke more unanimity in moral condemnation than the sexual abuse of children. The European Commission's determination to confront this evil is understandable and commendable. The proposal's supporters have highlighted a distressing reality — millions of images and videos of child sexual abuse circulating online, instances of horrific exploitation continuing unchecked, and investigative challenges posed by new technologies. They argue that extraordinary times call for extraordinary measures. In their view, the traditional tools have proven inadequate to curb online child abuse, and thus a bold new approach is warranted, leveraging Europe's regulatory power and technological capability to mandate comprehensive detection across digital communications. This perspective is grounded in a sincere desire to rescue children from harm and to prevent future abuse by leaving abusers "no place to hide".

However, as our exploration has shown, the *means* by which ChatControl seeks to achieve these ends are extraordinarily far-reaching. Implementing mass surveillance of private communications represents a paradigm shift in the relationship between citizens and the state (and indeed, between users and technology providers). It is not a change to be undertaken lightly or without unintended consequences. The analysis of the proposal's technical aspects revealed that it effectively demands either breaking encryption or installing state-mandated content scanners on personal devices — both options that pose severe security, privacy, and civil liberties concerns. The critiques from privacy experts, and human rights advocates are grounded in decades of experience with surveillance and abuse

of power, and in rigorous analysis of the limits of current technology. They warn that ChatControl, as envisioned, could generate a flood of false positives, divert law enforcement resources, enable new forms of attacks, and normalize a level of intrusion previously unseen in democratic societies. In trying to police the digital haystack, we risk burning it down or at least, charring the very fabric of trust and privacy that underpins the digital economy and personal relationships online.

Throughout this paper, a recurring theme has been the search for balance. Balancing child safety with privacy, technological capability with fundamental rights, enforcement efficiency with democratic oversight, and urgent action with careful restraint. The historical section illustrated that when faced with similar tensions (like generalized data retention), European institutions have ultimately leaned toward caution, striking down measures that overreached. The legislative journey of ChatControl so far – with pushback from Parliament committees, delays in the Council, and sharp criticism from judicial bodies – suggests that a straightforward implementation of the proposal in its original form is unlikely. There is growing recognition, even among some supporters, that modifications are needed to re-calibrate the proposal. The very notion of indiscriminate scanning may need to be abandoned in favor of more targeted, intelligence-led approaches. Indeed, if the final outcome of this legislative process is a system that focuses on suspects and high-risk scenarios (and is paired with strong safeguards and oversight), it may end up closer to what critics recommend than what was initially proposed. That would represent a victory for common sense and rights, achieved not by rejecting the goal of child protection but by insisting it be pursued in a manner consistent with European values.

The alternatives and solutions we discussed offer a roadmap for such a course correction. By investing in robust law enforcement cooperation, improving the processing of existing reports, taking down content at its sources, and bolstering preventive and educational measures, the EU can make significant strides against child abuse without resorting to mass surveillance. These measures may lack the shiny allure of an “AI panacea”, but they are grounded in proven practices and align with legal norms. Over time, steady improvements in these areas coupled with future technological innovations that might allow more nuanced interventions, could dramatically improve child safety online. Crucially, this can be done while still preserving the privacy of the vast majority of citizens who are law-abiding and whose communications merit confidentiality.

Ethically, the ChatControl debate has forced a societal reflection: How do we value privacy when weighed against protection from harm? The conclusion from this examination is that it is a false choice to say we must abandon one to secure the other. If anything, long-term child protection is best served in a society that values privacy and freedom, because abuse thrives in shadows of secrecy *and* in unchecked systems of power. A surveillance state can itself become a source of abuse, especially for the vulnerable. Thus, ensuring robust privacy and rule of law is part of protecting all citizens, including children.

In concluding, it is worth stepping back and recognizing that both the supporters and opponents of ChatControl share more common ground than it might appear. All agree that child sexual abuse is a heinous crime that must be combated. All acknowledge that doing nothing is not an option, and that technology companies have a responsibility to help address the misuse of their platforms. The difference lies in methodology and in foresight about consequences. The debate has been contentious, but it has also been productive in surfacing complex issues that need to be addressed as we chart the way forward.

The final shape of any EU policy in this domain should strive to integrate the valid insights of both sides. From the supporters, the urgency and resolve to protect children should be maintained; from the critics, the insistence on safeguarding fundamental rights and learning from technical realities must be heeded. The result could be a comprehensive strategy that elevates Europe as a leader in child protection *and* privacy. It might include a refined legislative measure (stripped of its most problematic elements), alongside funding programs for law enforcement training, victim support, public education, and privacy-respecting technological tool development.

In the end, we can measure the success. Are fewer children being abused and exploited? Are more offenders being brought to justice swiftly and fairly? And has this been achieved without dismantling

the liberties and security that define the European digital society? A thoughtful, multi-faceted approach as advocated in this expanded discussion offers a path to answer “yes” to all of these questions. As the EU moves forward, it has the opportunity to demonstrate that it is possible to *both* shield children from harm *and* uphold the rights of the populace – to deploy the tools of innovation and law in harmony with ethical principles. That is a vision of ChatControl’s goal achieved, but through methods we can live with, proudly and freely, for generations to come.

The lesson is philosophical as well as political. When fear governs technology, freedom contracts; our task is to build architectures of protection that do not demand obedience as their price.

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